Planning Ahead for Complicated Cable Access:

How do Low Sampling Rates Impact Earthquake Early Warning?

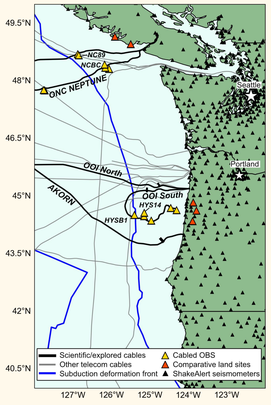
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Offshore data from either seismometers or DAS would greatly improve earthquake early warning performance in the case of an earthquake nucleating offshore, as expected for Cascadia.

However, on much of the U.S. West Coast, real-time streaming of offshore data is limited to 8 Hz. Workarounds to stream full-rate data (100+ Hz) may be possible, but data access would be greatly simplified if 8 Hz data is sufficient.

If the sensors nearest to a hypocenter only record at low sampling rates, how might that impact earthquake early warning performance?

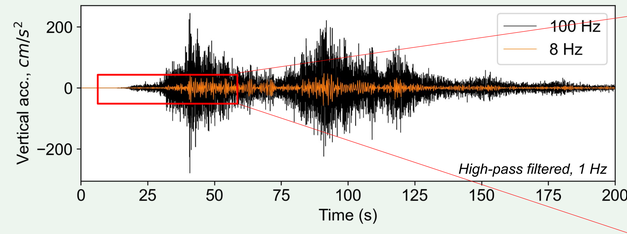


Existing Offshore Infrastructure

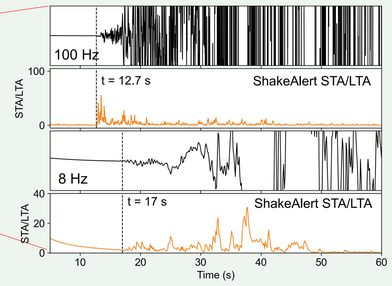
- Ocean Networks Canada NEPTUNE observatory: 5 broadband buried cabled ocean bottom seismometers (OBSs) on forearc, 2 on midplate
- Ocean Observatories Initiative (OOI) Regional Cabled Array: 2 broadband buried cabled OBSs with 4 planned for near-future deployment
- Distributed acoustic sensing (DAS) experiments have been carried out on the OOI and AKORN cables
- 18 other known private telecom cables span the subduction zone margin

Downsampling makes it harder to pick the initial P wave.

Tohoku M9.1 at 86 km on seismometer:

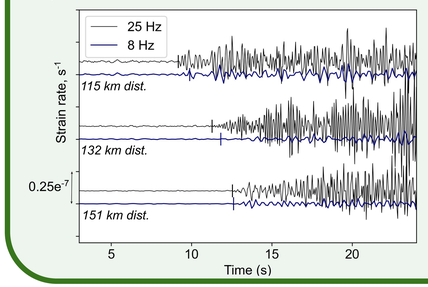


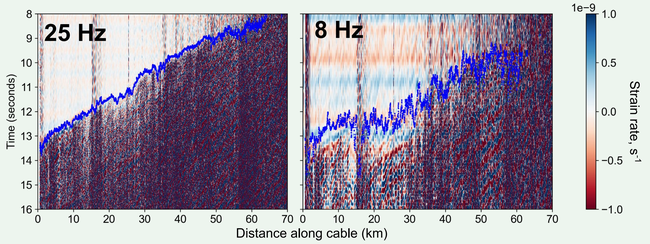
When using a ShakeAlert-style *P* wave triggering approach (EPIC's STA/LTA picker), the earliest the *P* wave is apparent on the 8 Hz data is several seconds behind the 100 Hz data:



Alaska M5 at 100+ km on DAS:

P wave is similarly more emergent & lower amplitude on lower rate DAS waveforms:



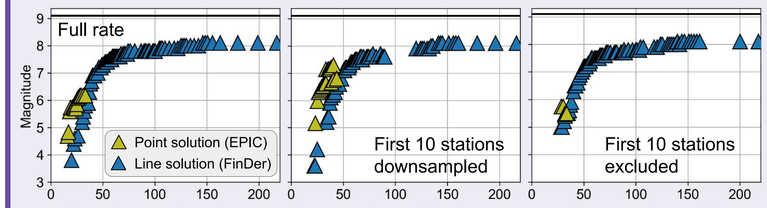


P wave picks using an ensemble machine learning approach (ticks on the left, blue points above) are more sparse, less precise, and tend to be delayed in lower rate data.

ShakeAlert

To approximate having low-rate data offshore (closest to hypocenter), we tested how downsampling the 10 stations closest to a hypocenter affects ShakeAlert performance. Performance varies, but larger earthquakes tend to still see better performance with downsampled data than without.

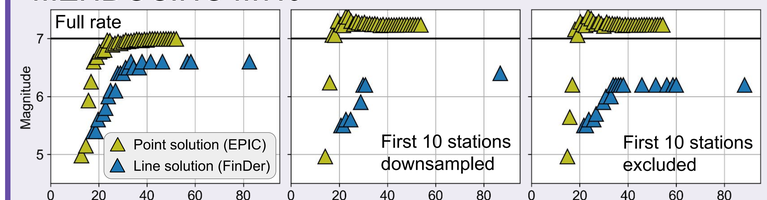
TOHOKU M9.1



- First solution at 16 s
- First solution at 22 s
- EPIC solutions are late and split
- First solution at 28 s

For Tohoku, downsampling the first 10 stations degrades EPIC performance significantly. When the first 10 stations are downsampled to 8 Hz, the first solution is 6 later, but 6 s faster than when the stations are excluded altogether.

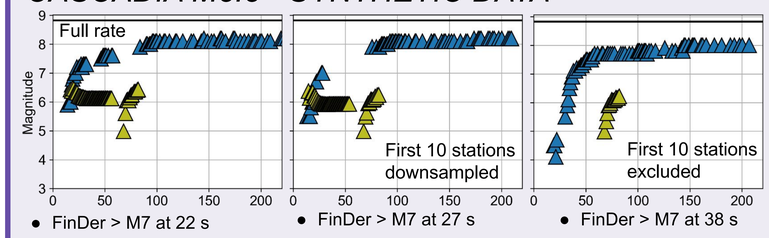
MENDOCINO M7.0



- First solution at 13 s
- FinDer mags peak at M6.5
- First solution at 14 s
- FinDer mags peak at M6.4
- First solution at 15 s
- FinDer mags peak at M6.2

For Mendocino, the downsampled stations (coastal, closest to offshore hypocenter) never generate triggers that are used by ShakeAlert. In this case, the EPIC solution is actually pulled further offshore, leading to higher magnitude estimates.

CASCADIA M8.8 - SYNTHETIC DATA



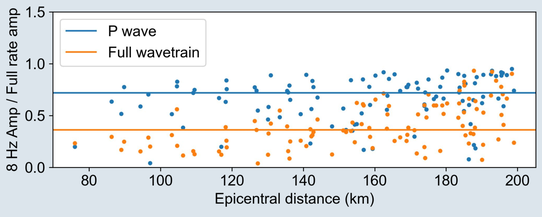
- FinDer > M7 at 22 s
- FinDer > M7 at 27 s
- FinDer > M7 at 38 s

For this synthetic Cascadia scenario, EPIC performance is not impacted by lower sampling rates; this may be due to the overall lower frequency content of the synthetic waveforms. Downsampling or excluding the offshore stations, however, slows the magnitude growth of the FinDer solution.

Downsampling reduces ground motion amplitudes…

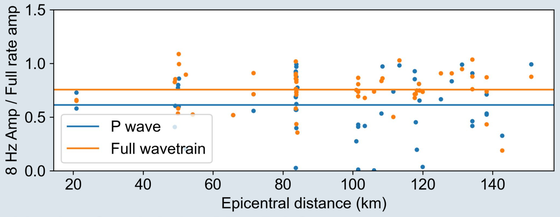
Tohoku 2011 M9.1

Full-rate (100 Hz) acc. Z channels downsampled to 8 Hz:
- P waves: 72% mean amplitude (blue, below)
- Maximum wavetrain peak: 36% mean amplitude (orange, below)



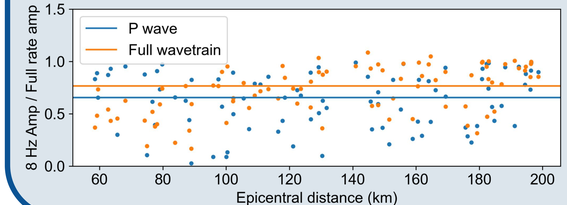
Alaska Cook Inlet M5: Seismometers + DAS

Full-rate (50-200 Hz) vel. Z channels downsampled to 8 Hz:
- P waves: 50% mean amplitude (blue, below)
- Maximum wavetrain peak: 72% mean amplitude (orange, below)



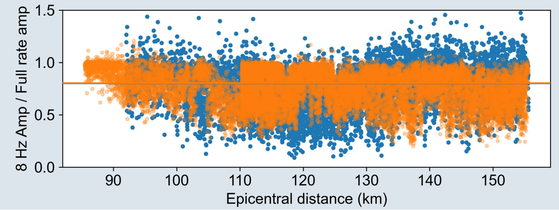
Mendocino 2024 M7.0

Full-rate (100-200 Hz) acc. Z channels downsampled to 8 Hz:
- P waves: 66% mean amplitude (blue, below)
- Maximum wavetrain peak: 76% mean amplitude (orange, below)



Alaska Cook Inlet M5: DAS

Full-rate (25 Hz) DAS channels downsampled to 8 Hz:
- P waves: 75% mean amplitude (blue, below)
- Maximum wavetrain peak: 76% mean amplitude (orange, below)



…which reduces earthquake magnitude estimates.

ShakeAlert's EPIC magnitude relationship:

$$M_w = 1.38*\log_{10}(P_D) + 1.23*\log_{10}(R) + 5.39$$

if we assume downsampled amplitudes only reflect a given percentage of full-rate amplitudes (40-90%, as observed), we can estimate associated magnitude reductions.

Assuming an M8 earthquake:

100% amplitude = Magnitude 8.0
90% amplitude = Magnitude 7.94
80% amplitude = Magnitude 7.87
70% amplitude = Magnitude 7.79
60% amplitude = Magnitude 7.69
50% amplitude = Magnitude 7.58

Take-aways

- The existing ShakeAlert algorithms would need to be substantially reworked to promote triggering on 8 Hz data. Even then, 8 Hz data lacks essential ground motion information.
- Smaller sampling rates may still be helpful in the case of larger earthquakes due to their higher energy at lower frequencies.